

Name: _____

Date: _____

RATIO, (DIVIDING IN A RATIO)

LO: I can divide a quantity into two or more parts in a given ratio.

Dividing in a ratio

To **divide in a ratio**, find the total number of **parts**, divide the quantity by the total parts to find one part, then multiply each ratio number by the value of one part.

Check: The parts must add back to the original total amount.

Quick examples

●	£60 in ratio 2:3 = £24 and £36	5 parts, 12 each
●	Total parts = sum of ratio numbers	$2 + 3 = 5$ parts
●	One part = total $\div$ sum of ratio	$60 \div 5 = 12$
●	£60 in 2:3 = £30 and £30	not equal shares

Key idea

Total parts = sum of ratio. One part = amount $\div$ total parts.
Multiply each ratio number by one part.

£60 in 2:3 one part = $60 \div 5 =$

12

Then $2 \times 12 = 24$ and $3 \times 12 = 36$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - two-part ratio

Share £80 in the ratio 3:5.

Total parts = $3 + 5 = 8$

One part = $80 \div 8 = £10$

3 parts = £30, 5 parts = £50

£30 and £50

Check: $30 + 50 = 80$.

Example 2 - three-part ratio

Divide 120 cm in the ratio 1:2:3.

Total parts = $1 + 2 + 3 = 6$

One part = $120 \div 6 = 20$ cm

Parts: 20, 40, 60 cm

20 cm, 40 cm, 60 cm

Check: $20 + 40 + 60 = 120$.

Example 3 - find one amount

Ali and Ben share money in ratio 4:7.
Ben gets £35. How much does Ali get?

Ben's 7 parts = £35

One part = $35 \div 7 = £5$

Ali's 4 parts = $4 \times 5 = £20$

Ali gets £20

Use the known amount to find one part first.

Example 4 - find the total

A drink is mixed in ratio 1:4 (concentrate:water). There are 200 ml of concentrate. Find the total.

1 part = 200 ml (concentrate)

4 parts water = $4 \times 200 = 800$ ml

Total = $200 + 800 = 1000$ ml

Total = 1000 ml

Find what one part equals from the given information.

YOUR TURN – PRACTICE (deliberate practice)

1. Two-part ratios

Divide each amount in the given ratio.

- a) £100 in ratio 2:3
- b) £72 in ratio 5:4
- c) 150 kg in ratio 1:4
- d) 200 m in ratio 3:7

2. Three-part ratios

Divide each amount in the given ratio.

- a) £60 in ratio 1:2:3
- b) 180 ml in ratio 2:3:4
- c) £240 in ratio 3:5:4
- d) 500 g in ratio 1:3:6

3. Finding one share

Find the missing amount.

- a) Ratio 2:5. Larger share is £40. Smaller share?
- b) Ratio 3:8. Smaller share is £15. Total?
- c) Ratio 1:3. Smaller share is 25 ml. Larger share?
- d) Ratio 4:9. Larger share is 54 kg. Total?

4. Word problems

Solve each problem.

- a) A 2 m rope is cut in ratio 3:5. Length of each piece?
- b) Paint mixed 2:5 (red:white). 350 ml needed. How much red?
- c) Prize money £500 shared 2:3. Winner gets?
- d) Concrete: cement:sand:gravel = 1:2:4. Need 2.1 tonnes total.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Total parts = sum of all ratio numbers ☐

Correct answer: _____

Reason: _____

b) Ratio 2:3 means divide into 2 equal and 3 equal parts ☐

Correct answer: _____

Reason: _____

c) £60 shared in ratio 2:3 gives £24 and £36 ☐

Correct answer: _____

Reason: _____

d) Ratio 3:4 means 3 out of 4 ☐

Correct answer: _____

Reason: _____

e) All parts added together should equal the original total ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Three friends share a £360 prize in the ratio of their ages: **12, 15, and 18**. How much does each person get? Simplify the ratio first.

Expression: _____

Simplified: _____

b) A rectangle has perimeter **84 cm**. The length and width are in the ratio **5:2**. Find the length, width, and area of the rectangle.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

